

Prolonged disease remission of the chronic immunological skin disorder hidradenitis suppurativa with adjunctive bacteriophage therapy

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Hidradenitis suppurativa (HS) is a chronic skin disease in which the acute development of noduli, abscesses and in a later stage fistulas remains difficult to control, despite a wide range of recommended treatment options including immunomodulatory biologicals, surgical intervention and antibiotics. Besides being painful, associated lesions have an extensive impact on the quality of life. While HS is primarily an immune-mediated inflammatory disease, the skin microbiota plays a key role in its pathogenesis with pathogenic bacterial colonization of lesions observed. Bacteriophages, the viruses of bacteria, are used to treat a 52-year-old female HS patient with re-current colonization of lesions with *Staphylococcus aureus*. Phage therapy results in complete removal of lesions with a flare-free period of six months, along with a substantial improvement in the patient's quality of life. The deployed experimental framework and the gained clinical experience will be valuable for future HS phage therapy research.

Hidradenitis suppurativa (HS), also called Verneuil's disease or acne inversa, is a chronic immune-mediated inflammatory disease (IMID) manifesting with painful skin lesions. The estimated global presence of HS ranges from 0.03% to 4% and impacts an individual's quality of life (QOL) to a great extent, with patients having

the lowest QOL scores of all cutaneous diseases^{1–3}. The pain accompanying nodule, abscess and fistula formation impairs daily activities and, in some cases, leads to complete inhibition of movement while sitting or lying down⁴. A combination of treatment strategies, including medication, surgery and lifestyle

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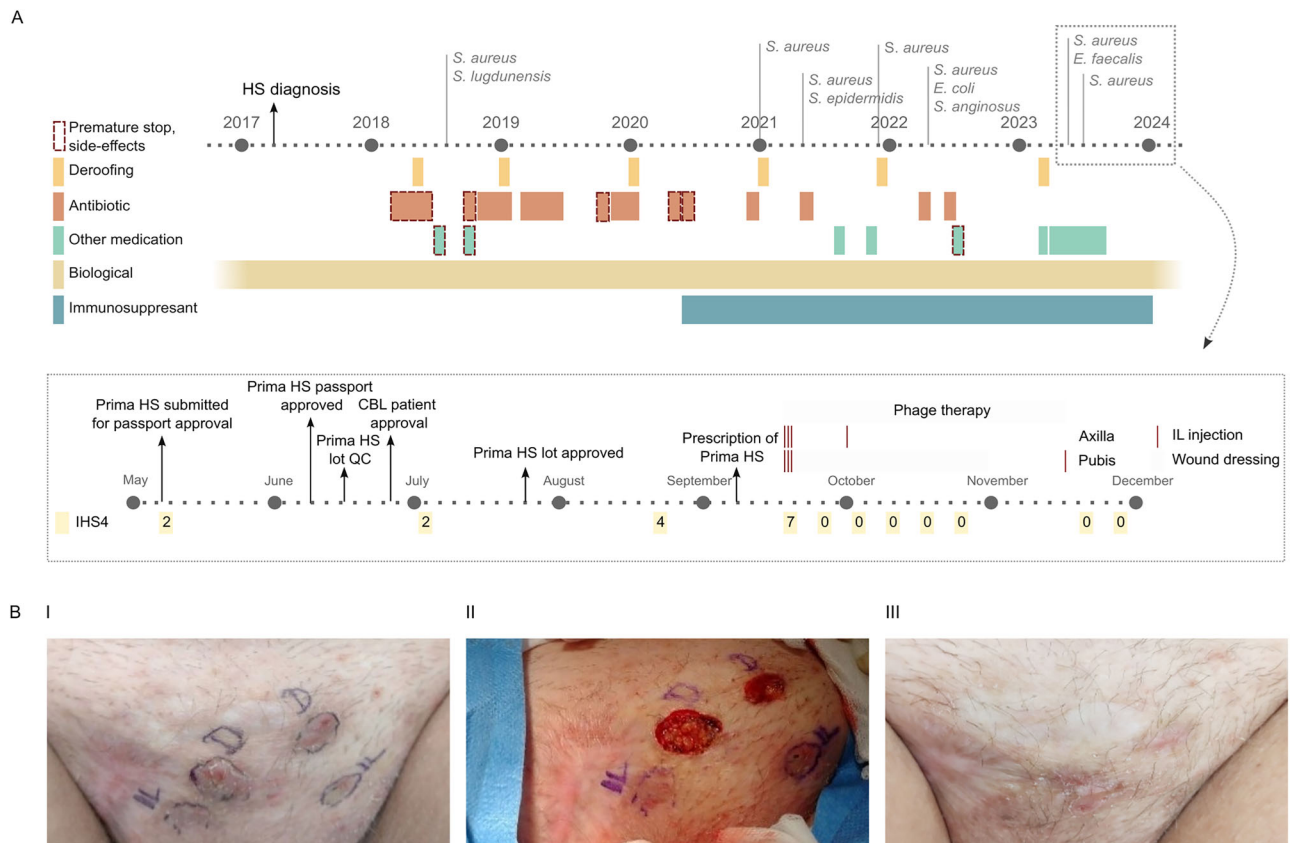


Fig. 1 | Patient disease history and phage therapy. **A** Patient timeline, indicating the bacterial species observed in the lesions, surgical (deroofing) and medical treatments (Antibiotics, Biological, Immunosuppressant and other medications recommended for HS) from HS diagnosis (2017) to 2024. Furthermore, the timeline

of phage therapy and its approvals is indicated together with the IHS4 score of the patient. Source data are provided as a Source Data file. **B** Pubis before therapy (I), after deroofing (II) and after phage therapy (III).

changes, are required, but often fail to control the disease adequately, leading to a therapeutic gap⁵.

Although an immune system dysregulation lies at the base of the disease, bacteria do play a profound role in this disease⁶. Previous studies have demonstrated that the skin microbiome of patients with HS significantly differs from that of healthy individuals, both in lesional and non-lesional areas, implicating microbial dysbiosis in disease initiation. This can facilitate the transition of commensal organisms to become pathogenic, thereby triggering immune activation and subsequent follicular occlusion^{7–9}. However, in certain individuals, an aberrant immune response to the symbiotic microbiota may also contribute to disease onset¹⁰. Beyond its role in disease initiation, bacterial colonization of established lesions further amplifies immune responses and inflammation, thereby boosting disease progression⁶.

Antibiotics remain the first-line treatment of HS due to their combined antimicrobial and immunomodulatory properties, and are frequently employed regardless of microbiological findings. They are used as monotherapy in mild cases and adjunctively in severe disease. Despite growing interest in biologic agents targeting cytokine pathways, antibiotics continue to play a critical role, particularly in managing acute flares¹¹. The prescription of antibiotics, however, is not an option for all HS patients, as this disease is known for its multitude of comorbidities, such as inflammatory bowel disease (e.g., Crohn's disease)¹². These patients often show gastrointestinal intolerance of antibiotic use.

Bacteriophage therapy has been proposed as a possible treatment strategy to regain bacterial equilibrium and prolong disease remission in HS patients¹³. Phages, the viruses of bacteria, are omnipresent across all ecosystems where they control bacterial populations¹⁴. Virulent,

strictly lytic phages infect and kill bacteria and due to their specificity (targeting a few strains to a considerable part of one single bacterial species) will not significantly affect the composition of the surrounding microbiota¹⁵. Moreover, phages are non-toxic to eukaryotic cells and are generally regarded as safe (GRAS), with serious side effects due to phage treatment rarely reported^{16,17}. Although the use of phages as a therapy to treat bacterial infections was already proposed at the beginning of the 20th century, the rise of antibiotics halted its development in Western medicine^{18–20}. In Eastern Europe, phage therapy was further developed and used over the past century²¹ and a limited number of HS patients were treated in Tbilisi, Georgia²². Although HS patients were reportedly successfully treated with phages, recent and documented scientific case reports are lacking^{23,24}. Furthermore, the mechanism by which phages affect disease development remains unknown, as besides their antibacterial activity, phages also possess immunomodulatory properties, which may be favorable for this chronic IMID²⁵.

In this case report, we present a 52-year-old female diagnosed with HS, with deep-seated lesions emerging in the groins and pubic region without formation of tunnels and fistulas, which are categorized as the frictional furuncle type of HS²⁶. Her long history of smoking and obesity (BMI of 37.18 kg/m²) both contribute to disease development and further imply this phenotype. The patient was also diagnosed with spondyloarthropathy and Crohn's disease, the latter often leading to gastrointestinal intolerance for recommended treatments of HS. As a result, the patient has a long record of medical and surgical treatments (Fig. 1A), which includes the application of antibiotics, retinoids, immunosuppressants, androgen antagonist, intralesional steroids and surgical deroofings between the diagnosis of HS in 2017 and the start

Table 1 | Phage host range on plate against *Saureus* strains isolated from the patient

Bacteria	Isolation site	Isolation date	ISP	Remus	Romulus	Prima HS
Host	–	–	1.00	1.00	1.00	1.00
StaAu009	Axilla	26.01.2021	1.5 × 10 ⁻⁵	0	0.36	0.94
StaAu035	Axilla	04.05.2021	1.5 × 10 ⁻⁵	0	0.19	1.2
StaAu120	Axilla	30.11.2021	1.5 × 10 ⁻⁵	0	0.31	1.2
StaAu121	Groin & genitals	30.11.2021	1.5 × 10 ⁻⁵	0	0.89	1.0

Efficiency of plating (EOP) of phages ISP (host *S. aureus* ATCC6538), Remus (host *S. aureus* PS47), Romulus (host *S. aureus* PS47) and Prima HS (host *S. aureus* PS47). For each bacterial isolate, the site and date of isolation is indicated.

Table 2 | Treatment, lesion count, IHS4 and PGA score over time

Treatment		Axilla		Pubis		Groin		Abdominal	Buttocks	IHS4	PGA
		N-N	I-N	N-N	I-N	A	I-N	N-N	I-N		
W1	IL, topical phage		1	–	2	2	–	–	–	7	3
W2	Topical phage	1		2	–		–	–	–	0	0
W3	IL, topical phage	1	–	–	–		–	–	–	0	0
W4	Topical phage	–	–	–	–		–	–	–	0	0
W5	Topical phage	–	–	–	–		–	–	–	0	0
W6	Topical phage	–	–	–	–		–	–	–	0	0
W10	IL phage	–	–	–	–		–	1	–	0	0
W11	/	–	–	–	–		–	–	–	0	0
W20	IL phage	–	–	–	–		1	1	–	1	3
W33	Clindamycin 2 × 300 mg	–	–	1	3		–	–	4	9	3
W50	/	–	–	1	–		–	–	1	3	3

The application of phages (intralesional (IL) or topical) and antibiotics, as well as the lesion count at the axilla, pubis, groin and abdominal fold, including non-inflammatory noduli (N-N), inflammatory noduli (I-N) and abscesses (A) are shown.

of phage therapy. In addition, she is on a stable weekly regimen of adalimumab and methotrexate. Although the chronic course of the disease is suppressed maximally with biological and combination therapy, acute flares present regularly and are very painful. Specifically, six months before phage therapy, a deroofting was performed, but acute flares again developed within three months. These flares were treated with intralesional steroids but relapsed every two weeks. Since her diagnosis of HS in 2017, the opportunistic pathogen *Staphylococcus aureus* was identified at different timepoints and at different stages of the disease in the biopsy of inflammatory nodules, in pus from abscesses and in swabs of healthy skin. In September 2023, phage therapy was initiated under the magistral phage framework to treat the *S. aureus* colonization²⁷ in acute flares. This first case led to the establishment of a clinical framework for the application of phages in HS.

Results

Prima HS, an evolved phage that targets HS *S. aureus* strains

Among the *S. aureus* phages in our collection, including ISP, which was previously used in various other clinical case studies^{17,28}, only Romulus²⁹, a member of the *Silviavirus* genus, showed potent lytic activity against the *S. aureus* strains isolated on different time points from the considered HS patient, who was eligible for phage therapy (Table 1).

Phage Romulus was further adapted using an adapted Appelmans’ method, resulting in the phage Prima HS (Romulus(Tyr311Ser)), which showed increased activity on all patient’s isolates as shown by an increase in efficiency of plating (EOP) (Table 1). Prima HS could inhibit in vitro growth of StaAu009 for 20 h (MOI of 1), whereafter bacterial growth was observed (Supplementary Fig. 2). Interestingly, Romulus also showed activity on other HS patient bacterial strains (5/8), with Prima HS showing increased activity for most of these strains (Supplementary Table 1). Whole genome sequencing (WGS) and

comparative analysis of Prima HS to the original Romulus genome (NC_020877) identified one single nucleotide polymorphism (SNP) leading to a missense mutation (Tyr311Ser) in the open reading frame (ORF) 42, identified as a structural protein within the tail protein complex of Romulus. This new HS pre-adapted phage Prima HS received a “green genetic passport” from Sciensano, allowing its subsequent use as a seed lot for API productions and ultimately its therapeutic use as an ingredient in a magistral preparation for the selected patient.

Phage therapy leads to successful disease relief of acute lesions

This patient had moderate HS (IHS4 score of 7) and standard treatment (antibiotics and biologicals in combination with antiseptics) failed or had contra-indications. As a phage that could efficiently infect the patient’s bacterial isolates was available and informed consent was signed, the patient was eligible for phage therapy following the PHA-GEFORCE inclusion criteria. Two inflammatory noduli and two incipient abscesses in the pubical region, and one inflammatory nodule at the axilla were identified as treatment sites by the treating physician (Table 2 and Fig. 1B–I). Sonography confirmed the presence of these deep-seated, delineated lesions.

No changes in heart rate, blood pressure or temperature were observed before and after phage application. Blood tests also showed no abnormalities, and no adverse reactions were observed. One week after the start of phage therapy, inflammation subsided in the injected noduli, and two weeks after therapy start both noduli at the pubis were completely gone. As the nodule at the axilla remained still palpable, although non-inflammatory, an additional intralesional injection of phage was carried out, and one week later, this nodule was no longer present. After wound healing, no HS lesions were observed at the location of the deroofting. Removal of all five lesions was also confirmed by sonography, in which only scar tissue was observed at the deroofted locations (Fig. 1B–III). The IHS4 score remained zero for three

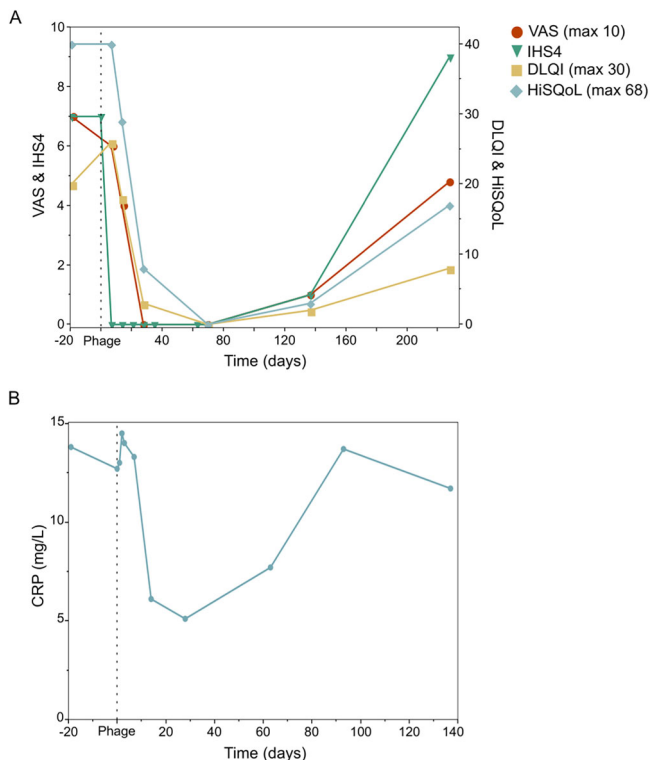


Fig. 2 | Clinical scores. A The Visual Analog Scale (VAS, red circle) and International Hidradenitis Suppurativa Severity Score System (IHS4, green triangle) scores (left y-axis) and Dermatological Life Quality Index (DLQI, yellow square) and Hidradenitis Suppurativa Quality of Life (HiSQoL, blue diamond) scores (right y-axis) before and after the start of phage therapy (day 0). Source data are provided as a Source Data file. **B** C-reactive proteins (CRP) levels in the blood before and after the start of phage therapy (day 0). Source data are provided as a Source Data file.

months after the phage treatment course of six weeks (W20, Fig. 2A). In this period, a new, small non-inflammatory nodule developed at W10 at the abdominal fold, but after intralesional phage injection, this nodule disappeared after one week. Development of both an inflammatory and non-inflammatory nodule at W20 in respectively the groin and abdominal fold, led to an IHS4 score of 1, which again subsided after phage injection. Six months after phage therapy (W33), the IHS4 score increased substantially to 9 due to the emergence of new noduli and an abscess in the phage-treated region (pubis), together with noduli at the buttocks. As phage treatment was finished, according to the protocol, treatment with clindamycin was started, but already after one month, the patient lowered the dose due to gastrointestinal side effects, leading to insufficient treatment.

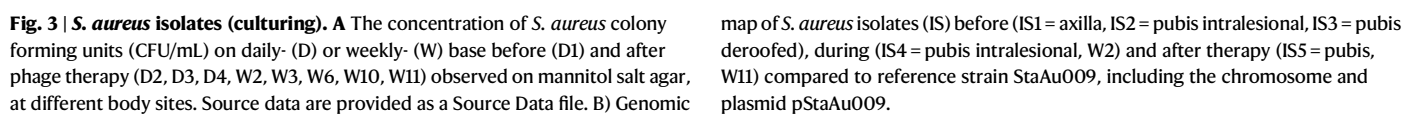
In addition to the clinical course of the disease, a large positive impact on the quality of life and mental health of the patient was observed (Fig. 2A) as both the DLQI and HiSQoL went to zero after complete healing of the wounds. The pain indicated by the patient on the VAS also decreased to zero. These scores remained zero until the development of new lesions, but only showed a minor increase. Furthermore, the level of C-reactive protein (CRP), indicating the level of systemic inflammation, dropped from 14.6 mg/L to 5.1 mg/L during the phage application (Fig. 2B), which is more close to the normal range (CRP ≤ 5.0 mg/L). No effect of phage therapy was visible in the TNF- α serum levels over time (13.9 pg/mL $\pm$ 1.75 pg/mL, Supplementary Table 2) and the concentration of IL-17 in serum remained below the detection limit of 4 pg/mL. No significant phage immune neutralization was detected between the different timepoints (Wilcoxon Signed-Rank, mean equal to zero, p -value > 0.05) (Supplementary Fig. 3).

Microbiological follow-up indicates that clonal diversity does not impact therapy

At the start of this study, *S. aureus* strain StaAu009 was isolated from the axilla and further used as a reference isolate for this patient. WGS using both short-read and long-read sequencing technologies assigned the isolate to Sequence Type (ST)7, a prevalent *S. aureus* clone. One plasmid was identified (pStaAu009), which harbored antibiotic resistance genes *fusB* and *blaZ*, mediating resistance against fusidic acid and penicillin, respectively. Fusidic acid resistance was confirmed by the antibiogram (Supplementary Table 3). Virulence genes were chromosomally encoded and included genes for toxins (*hlgA*, *hlgB*, *hlgC*, *lukD*, *lukE*, *sep*), exoenzymes (*aur*, *splA*, *splB* and *splE*) and host-immune genes (*sak* and *scn*).

Already after one day of phage treatment, no *S. aureus* could be detected in skin swabs of the treated lesion in the axilla by culturing, and long-term removal of the pathogen was observed (Fig. 3A). In contrast, phage therapy did not eradicate this bacterium from the pubic area, as it was still detected three months after the start of therapy. However, a reduction in titer was observed. *S. aureus* strains isolated immediately before (W1: IS1, IS2, IS3), during (W2: IS4) and after therapy (W11: IS5) were clonal compared to StaAu009 and comparative analysis revealed an increase in SNPs post-phage treatment isolates (Fig. 3B). Interestingly, one deletion was identified in isolate IS5 (W11), in the gene *clfA*, encoding clumping factor A, a virulence factor important for invasive *S. aureus* disease. No changes in antibiotic resistance were observed, and all isolates were still susceptible to phage Prima HS (Supplementary Table 3).

16S rRNA profiling of the bacterial community was successfully carried out on lesional and non-lesional skin swabs taken before, during and after treatment. Multivariate distance-based redundancy analysis (dbRDA, Bray-Curtis) on the samples indicated that the sampling timepoint ($R^2_{adj} = 0.176$, BH corrected $p = 0.002$), the therapy zone ($R^2_{adj} = 0.140$, BH corrected $p = 0.002$) and the sampling location ($R^2_{adj} = 0.050$, BH corrected $p = 0.002$) had a significant effect on the bacterial composition, with a total combined effect size of 36.6%. Due to this significant effect of the therapy zone (axilla or pubis), bacterial diversity analysis was also carried out per therapy zone. In the axilla, no significant differential abundance of a bacterial species was observed. Nevertheless, the sampling time point did have a significant effect on the bacterial composition (univariate dbRDA, Bray-Curtis; $R^2_{adj} = 0.268$, BH corrected $p = 0.003$). The number of observed species was on average 75 ± 17 in healthy skin, while this was lower in the noduli (54 ± 14 ; $p = 0.0091$) and after disappearance of noduli (50 ± 4 ; $p = 0.307$) (Wilcoxon each pair). Healthy skin next to the lesion and on the mirroring site of the body did not have a different number of observed species (Fig. 4A). No significant difference in Shannon alpha-diversity between timepoints, sampling location or skin status was present (Wilcoxon test). In lesional skin, a decrease in abundance of *S. aureus* was observed with 15.3% relative abundance before therapy (D1) and no detection of *S. aureus* five weeks after start of therapy (W5). In the pubic area, both timepoint ($R^2_{adj} = 0.352$, BH corrected $p = 0.002$) and sampling location ($R^2_{adj} = 0.092$, BH corrected $p = 0.002$) had a significant effect on the bacterial composition as examined with a multivariate dbRDA (Bray-Curtis) with a combined effect of 44.4%. Differential abundance analysis confirms that the relative abundance of *S. aureus* decreases over time (ALDEx2, Benjamini-Hochberg corrected $p < 0.05$), with a relative abundance of 0.52 ± 0.14 before therapy (D1) and 0.0023 ± 0.00085 five weeks after the start of therapy (W5) (Fig. 4B). No differential abundance of this pathogen was detected between sampling locations or between samples with different skin status. No difference in alpha diversity (Observed species and Shannon) was observed between timepoints, sampling location or skin status (Wilcoxon test).



A pilot HS patient was selected based on her long medical history, recurrent colonization of the lesions with *S. aureus*, and the availability of a potent phage against the bacterial isolate. The HS patient showed colonization of *S. aureus* in her lesions since her diagnosis in 2017, and phage Romulus could infect all *S. aureus* strains recovered from this patient over the time span of one year before treatment. Directed evolution of phage Romulus against HS *S. aureus* isolates resulted in a new phage isolate named Prima HS, which emerged to exhibit higher infectivity linked to one single missense mutation in a structural protein of the phage's tail complex. In Belgium, phages can be prescribed by medical doctors and delivered as magistral preparations under the direct responsibility of doctors and pharmacists. Upon quality and safety assessment by the independent Scientific Institute of Public Health (rebranded as Sciensano), according to the quality control requirements included in the phage API monograph, this phage received the necessary certificates that allow its therapeutic use as part of magistral preparations²⁷. This means that upon prescription by the physician, phage Prima HS can be used in phage therapy against *S. aureus*. Since half of the HS *S. aureus* strains tested from different patients (8/16) were susceptible to phage Prima, this treatment can potentially be implemented for additional HS patients in the future.

Based on the PHAGEFORCE protocol, a preliminary clinical framework for phage therapy for HS was established and applied³¹. Phage therapy successfully alleviated acute flares, showing not only a drop in the IHS4 score from seven to zero after one week, but also complete cessation of pain and significant improvement in quality-of-life scores following the healing of surgical wounds. Phage therapy was able to induce a remission period of about six months before inflammatory lesions returned in one of the phage-treated areas. This appears comparable to antibiotic regimens, which achieve flare-free periods of five months on average after a 10-week course of oral clindamycin and rifampicin³², but without the toxicity, collateral damage to the microbiota and risk of increasing antibiotic resistance. To date, 11 months after treatment, no additional lesions have been observed in the axilla. Moreover, injection of phages in noduli that developed after the initial six-week treatment course, in regions not originally treated with phages, always resulted in complete removal of the lesion after one week, showing that early phage treatment can prevent the progression to more severe lesions like abscesses or fistulas.

S. aureus is considered a pathogen, but is also part of the commensal skin microbiome³⁴. A high concentration of *S. aureus* was observed in HS lesions pre-treatment. At the axilla, no *S. aureus* could be detected using culturing after disappearance of the lesion and also

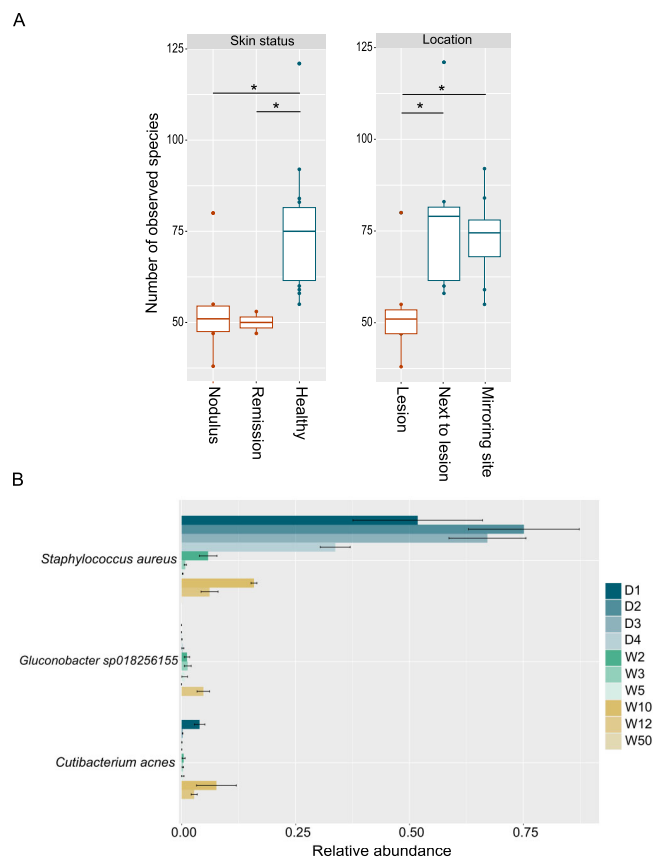


Fig. 4 | 16S rRNA profiling of the bacterial communities. **A** Number of observed species (alpha diversity) at the axilla, samples grouped by skin status and location. In each group, biological different samples (skin swabs) were used (nodulus: 6, remission: 2, healthy: 15; lesion: 8, next to lesion: 7, mirroring site: 8). Boxplots indicate the median (central line) and the 25th and 75th percentiles (box edges). Whiskers extend to the minimum and maximum values within the interquartile range. Only data points outside the 25th and 75th percentiles are shown individually. Significant differences were evaluated with a two-sided Wilcoxon test, * significant p -values: Nodulus – Healthy (0.0091), Remission – Healthy (0.0307), Lesion – Next to lesion (0.0076), Lesion–Mirroring site (0.0085). Source data are provided as a Source Data file. **B** Differential abundance analysis (DAA) of pubic samples over different timepoints (D: day, W: week). The mean relative abundance with standard error of the mean of species identified as significantly differentially abundant are shown. Each bar shows data from three biological samples. Significantly differentially abundant species were identified with a two-sided Benjamini–Hochberg corrected Wilcoxon test, * p -value < 0.05. Source data are provided as a Source Data file.

16S rRNA profiling showed no abundance of *S. aureus* five weeks after start of therapy. However, this did not lead to an increased number of bacterial species to level the numbers observed in healthy skin. Both culturing and 16S rRNA bacterial community analysis showed that phage therapy did not completely eradicate *S. aureus* at the pubis, but it likely decreased the abundance of this opportunistic pathogen to a more normal level. Phages in nature control bacterial communities, keeping them in check. Taking advantage of this trait may be preferential in treatments where the cause or exacerbation of disease is due to an overabundance of a normal member of the microbiota. The use of phages could potentially be synergistic with recommended medicine for HS, as it was previously already hypothesized that a loss of response to adalimumab could be associated with nasal colonization of *S. aureus*³⁵. To date, it is unclear if phages lead to disease remission due to their antibacterial effects alone, or if their capability to modulate the immune system also has an influence. No effect on the TNF- α serum levels were detected, but the CRP level did decrease

during phage therapy, which could also be explained by the bacterial clearance.

All bacterial isolates before, during and after phage application were clonal and did not show a change in susceptibility to phage Prima HS. Also, no change in antibiotic resistance occurred during therapy. Moreover, little to no immune neutralization of phages was detected. Thus, it seems that the six-week treatment regimen did not lead to therapy resistance attributable to bacterial resistance or antibody development against the phage. Although phages will not be applied as a continuous treatment, further follow-up of resistance development remains necessary, as re-application of phage will be required during future flare-ups. However, as the natural controllers of bacteria, phages will be able to co-evolve with their host. As with many chronic diseases, it is expected that HS patients will require a life-long treatment regimen, opening the door towards long-term application of phages, which should be accompanied by microbiological and immunological assessments.

This study presents a single-patient case report, and as such, its findings should be interpreted with caution. The absence of a control group and the complexity of the patient's treatment make it difficult to isolate the specific contribution of phage therapy to the observed clinical outcome. While the temporal association between phage application and lesion resolution is compelling, further studies involving larger cohorts and controlled treatment settings are necessary to validate the efficacy and immunomodulatory potential of phage therapy in HS. Nonetheless, this case provides valuable preliminary insights, as well as a scalable clinical and regulatory framework for future implementation.

In conclusion, a 52-year-old female patient diagnosed with the chronic immune disorder HS was successfully treated with phage therapy, resulting in a flare-free period of six months in the treated area. Application of pre-adapted *S. aureus* phage Prima HS did not only result in complete removal of all lesions, but also in a significant increase in quality of life. A clinical framework for phage therapy for HS was established, in which intralesional application of phages showed no adverse reactions. Although no resistance against the therapy emerged, close follow-up will be necessary as re-application of the therapy will be required to control this chronic disease, presenting a path towards the long-term, intermittent application of phages.

Methods

Study design and ethical approval

This study was approved by the Ethics Committee Research UZ/KU Leuven (S64854) as an amendment to the PHAGEFORCE study³¹. In this study, individual cases eligible for bacteriophage therapy are assessed and approved by the coordination group for bacteriophage therapy in Leuven (CBL). A dedicated HS phage therapy protocol was established based on the available PHAGEFORCE protocol and in concertation with the CBL. In here, the administration protocol was based on the experience of phage therapy experts at the Eliava Institute (Georgia) and Queen Astrid Military Hospital (Belgium) (Supplementary Methods – Phageforce protocol for Hidradenitis Suppurativa). Sampling of HS lesions (without phage therapy) was carried out under the DALIHS biobank (S62301), for which all individuals signed an informed consent to participate and for the publication of data. No financial compensation was provided to the participant for participation in this study. The sex of the participant was determined based on self-report during clinical intake. As this is a single-patient case report, no sex or gender-based comparative analysis was conducted. Therefore, disaggregation of data by sex or gender was not applicable.

Bacterial sampling, identification and antibiogram

Skin lesions were sampled (eSwab®, Copan), and bacteria were cultured by streaking the swabs on blood agar (Biomérieux), followed by incubation at 37 °C. Identification of the bacterial taxa was carried out

by matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF) mass spectrometry (Bruker UK Ltd, MBT compass library Revision L). After bacterial purification by replating of single colonies, antibiotic susceptibility was examined and categorized according to the EUCAST guidelines³⁶. The concentration of Colony Forming Units (CFU) was determined by preparing tenfold dilutions of swab liquid in PBS (137 mM NaCl, 2.7 mM KCl, 10 mM Na₂HPO₄, 1.8 mM KH₂PO₄) and plating 50 µL of each dilution on mannitol salt agar (mannitol salt 2, Biomérieux). The number of colonies in the initial sample was determined after overnight incubation at 37 °C.

Phage efficiency of plating and time-kill curves

The efficiency of plating (EOP) of phages was determined in lysogeny broth (LB) medium, as described by Kutter (2009)³⁷. The activity of the phages against the bacteria and development of bacterial phage resistance were examined by determining the 24 h time-kill curves of the phages, at a multiplicity of infection (MOI) of 1, using the Omnilog[®] system (Biolog, Hayward, CA, USA), as described by Merabishvili et al. (2024)³⁸. The reduction of the tetrazolium dye associated with bacterial respiration was monitored and recorded every 15 min at 37 °C. Each growth experiment was performed using three biological replicates. Omnilog[®] data analysis software was used to analyze the data, which was visualized using JMP Pro 17.

Pre-adaptation of phage Romulus and approval of phage Prima HS

Phage Romulus (vB_SauM_Romulus) was pre-adapted using a modified Appelmans' protocol^{38,39}. Two HS clinical isolates of *S. aureus*, StaAu008 or StaAu009, were used to obtain increased lytic activity and delayed resistance emergence. Pre-adaptation was performed on both bacterial isolates separately using an Omnilog[®] system (Biolog)³⁸. A tenfold dilution series of phage Romulus was mixed with bacteria and incubated overnight at 37 °C while shaking. The wells with the lowest MOI showing the longest bacterial growth inhibition were selected, mixed and filtered. The resulting phage lysate was used for the next round. In total, five rounds of adaptation were carried out whereafter the titer of the final lysate (expressed in plaque-forming units (PFU)/mL) was determined using the double agar overlay method⁴⁰. The phage variant adapted on the StaAu008 strain showed a longer inhibition of bacterial growth on this strain (Supplementary Fig. 1) and higher EOP on both HS strains compared to the host of Romulus (*S. aureus* PS47) (Table 1 and Supplementary Table 1). As a consequence, this isolate was selected for subsequent steps of propagation, purification and sequencing. The adapted phage, now called Prima HS, was produced as an active pharmaceutical ingredient (API) at the Queen Astrid Military Hospital (Brussels, Belgium) and submitted to Sciensano (Brussels, Belgium), formerly known as the Scientific Institute of Public Health, which checked the quality and safety of the phage in accordance with the requirements specified in the Belgian bacteriophage API monograph²⁷. This master seed of phage Prima HS and its production host (StaAu008) received a "green genetic passport" (06/06/2023), which meant that this produced phage was allowed to enter the phage bank for further use as a starter for phage API production. The subsequently produced Prima HS API batch passed Sciensano's quality control and was thus ready for inclusion in phage magistral preparations, prepared in a hospital pharmacy, upon prescription by a physician.

Treatment protocol of phage therapy for HS

Patient inclusion criteria. Patients diagnosed with moderate to severe HS (International Hidradenitis Suppurativa Severity Score System (IHS4) ≥ 4 ⁴¹) are eligible for phage therapy when the treatment with antibiotics and biologicals, both in combination with antiseptic treatment, failed or had contraindications, or when no treatment options are available for the patient. In specific for the antibiotic treatment, this

comprises multiple courses of antibiotic treatment with different types of antibiotics, including tetracyclines for at least four months and a combination of clindamycin (2 × 300 mg) and rifampicin (2 × 300 mg) for at least ten weeks.

When a phage targeting the bacterial strain colonizing the lesion is available, phages can be applied upon prescription by the treating physician, and minding the patient's informed consent.

Phage application. The patient was hospitalized for a four-day period to closely monitor the effects of the phage application. The hospital pharmacy prepared the phage solution by diluting phage Prima HS to a final concentration of 10⁷ PFU/ml in 0.9% NaCl and pH 5.5, 0.5 ml and 6 ml solution at 10⁷ PFU/ml were respectively compounded in 1 ml polycarbonate syringes (BD, ref. 309628) and 10 ml polypropylene syringes (BD, ref. 300912) and stored at 4 °C before administration. Stability of the phage solution in prefilled syringes was set at 7 days based on a recent stability study⁴². Depending on the type of lesion, phage solution (10⁷ PFU/mL) was applied intralesional (for noduli) or topical (for derroofed abscesses). During the period of phage treatment, the patient continued chronic treatments (weekly regimen of adalimumab and methotrexate) but did not use a topical antiseptic wash. Before each phage application, the skin was cleaned with physiological water. Inflammatory noduli were injected with 0.5 mL phage solution three days in a row. Abscesses were derroofed prior to phage application. Twice a day, the open wounds were irrigated with 2 mL of phage solution and covered with a dressing soaked in 4 mL of phage solution. This treatment course was maintained until the wounds were completely healed, i.e., six weeks. Weekly follow-up consultations were carried out, during which new phage solutions were provided to the patient for topical phage administration at home, and intralesional injections were repeated when the noduli were still palpable or new noduli emerged.

Safety and efficacy assessment. Before, during and after phage application (one week, one month, three months and six months after phage therapy), the patient's clinical parameters (heart rate, blood pressure and temperature) were followed. In addition, blood and lesion samples were collected. Blood tests were carried out, including complete blood cell count, basic metabolic panels, inflammatory parameters, lactic acid, creatine kinase and liver function tests. Serum samples were prepared by letting whole blood clot for at least 30 min prior to centrifugation for 10 min at 2000 × g (Megafuge 1.0, Heraeus), after which the serum was aliquoted per 0.5 mL and stored at −80 °C. The level of tumor necrosis factor alpha (TNF-α) and interleukin 17 (IL-17) in the serum was determined with respectively the TNF-α human uncoated ELISA kit (ThermoFisher Scientific) and the IL-17A (homodimer) human uncoated ELISA kit (ThermoFisher Scientific), in quadruplicate. Swabs of the lesions were taken in triplicate to determine the concentration of the targeted bacteria (*S. aureus*) and the applied phage (Prima HS). Additional swabs were taken next to the lesion and on the same location, but on the other side of the body, were collected to assess the changes in bacteriome composition. HS assessment was carried out considering the number and types of lesions to determine activity and severity scores, such as Hurley stage, IHS4 and Physician Global Assessment (PGA), and to measure clinical outcome parameters such as drop in IHS4^{41,43}. Patient Reported Outcomes (PROs) were obtained through Visual Analog Scale (VAS), Dermatological Life Quality Index (DLQI) and Hidradenitis Suppurativa Quality of Life (HSQoL) questionnaires^{43,44}.

Phage immune neutralization

Serum was diluted 1:100 in 0.9% NaCl solution, and 900 µL of the diluted serum was mixed with 100 µL of phage solution (10⁷ PFU/mL). The activity of the phage was determined, directly after mixing of phage and serum and after 30 min of incubation at 37 °C, using the

double agar overlay method⁴⁰. For this, tenfold serial dilutions of the phage-serum mix in 0.9% NaCl were prepared (in duplicate) and 10 µL of the dilutions were spotted on a double agar overlay (LB agar plate (1.4% agar) with 4 mL LB soft agar layer (0.4%) containing 200 µL bacterial overnight culture). The percentage of phage neutralization was determined by dividing the average PFU/mL after 30 min incubation by the initial average titer^{40,45}. For each serum sample, the experiment was carried out in triplicate. Statistical analysis was carried out using JMP Pro 17, for which a Wilcoxon Signed-Rank test was applied to examine if the mean was equal to zero. A significance level of 0.05 was used.

Genomic analysis

The bacterial genome (StaAu009) was sequenced as previously described⁴⁶. Briefly, total genomic DNA was extracted from a single strain overnight liquid culture grown in LB (DNeasy UltraClean Microbial kit, Qiagen), prepared using the Nextera Flex library kit (Illumina) and sequenced on an Illumina Miniseq. In addition, nanopore long-read sequencing using the rapid barcoding kit SQK-RBK114.24 was performed on a MinION equipped with an R10.4.1 flow cell (Oxford Nanopore Technologies). Basecalling was carried out with Guppy (v6.3.8, Oxford Nanopore Technologies). Quality of the reads and trimming was performed for Illumina reads (Fastqc (v0.12.1, <https://github.com/s-andrews/FastQC>) and Trimmomatic (v0.39)⁴⁷) and nanopore reads (Nanoplot (v1.41.0, <https://github.com/wdecoster/NanoPlot>) and Porechop (v0.2.4, <https://github.com/rrwick/Porechop>)). A hybrid assembly of the StaAu009 genome was constructed using Unicycler (v0.5.0)⁴⁸ and annotated using Prokka (v1.14.6)⁴⁹. Virulence genes were identified using VirulenceFinder 2.0⁵⁰, antibiotic resistance genes with ABRicate (v1.0.1, <https://github.com/tseemann/abricate>), and multilocus sequencing typing (MLST) analysis was performed using MLST (v2.22.0)⁵¹. Further annotations were performed using EggNOG-mapper (v2.1.8)⁵², PHASTER⁵³, mobileOG-db (v1.1.2)⁵⁴, GC% (readGC v5.0.1, <https://github.com/angelovangel/readGC>), and GC calc (v1.0.0, <https://github.com/WenchaoLin/GCcalc>). Genomic graphs were generated using Circos⁵⁵. Subsequent *S. aureus* isolates of this patient (IS1, 2, 3, 4 and 5) along with other HS isolates (StaAu008, 131, 053, 067, 092, 190, 102, and 191) and the genome of Prima HS were sequenced using the Illumina MiniSeq platform alone with reads processed as described above and short-read assembly carried out using Unicycler (v0.5.0)⁴⁸. Comparative analysis was performed using Snippy (v4.6.0, <https://github.com/tseemann/snippy>). The genome of Romulus available on NCBI was used as a reference (GCA_000907055.1) for phage comparison, and the genome of StaAu009 was used as a reference for bacterial comparison.

16S rRNA gene profiling

The DNA of the swabs was extracted with the DNeasy UltraClean Microbial kit (Qiagen) according to the manufacturers protocol, although the elution step was performed with 20 µL buffer instead of 50 µL. 16S rRNA genes were barcoded and amplified using the Kinnex 16S rRNA library workflow of PacBio, whereafter they were sequenced using PacBio Sequel IIe. In total, 66 samples were processed (Source Data File). Processing of the sequencing reads into operational taxonomic units (OTUs) and taxonomic classification was achieved using the HiFi-16S-workflow v0.7 (GitHub - PacificBiosciences/HiFi-16S-workflow). 16S rRNA analysis, statistical tests and visualization were performed at the species-level with R version 4.2.2⁵⁶ by making use of the *qiime2R*⁵⁷, *phyloseq*⁵⁸ and *ggplot2*⁵⁹ packages. Differentially abundant species were identified with *ALDEx2*⁶⁰ (Wilcoxon test, Benjamini-Hochberg (BH) corrected). Alpha-diversity, including observed taxa and Shannon diversity, was determined with *phyloseq* and the Wilcoxon test was used to examine significant differences. The contribution of metadata variables on the microbial composition was determined by distance-based redundancy analysis (dbRDA). The

independent contribution was determined with the capscale function of *vegan*⁶¹, for which a BH correction for multiple testing was applied. Variables that showed a significant contribution to the ordination were further examined in a stepwise dbRDA (*ordiR2step* of *vegan*) to determine the cumulative contribution. For all statistical tests, a significance level of 0.05 was used.

Statistics and reproducibility

This study is a single-patient case report and does not include statistical comparisons between groups. No statistical method was used to predetermine sample size. No data were excluded from the analyses. The experiments were not randomized, and the investigators were not blinded to allocation during experiments or outcome assessment.

To ensure reproducibility, all experimental procedures were conducted using validated and standardized protocols. Phage efficacy testing with time-kill curves was performed in triplicate using the OmniLog® system. Phage immune neutralization assays were conducted in triplicate per serum sample, and a Wilcoxon Signed-Rank test was applied to assess if the mean was equal to zero. ELISA measurements for TNF-α and IL-17 were carried out in quadruplicate using commercial kits. In addition, 16S rRNA gene profiling was performed on 66 samples using the HiFi-16S workflow, with statistical analysis conducted in R using the *phyloseq*, *qiime2R*, and *ALDEx2* packages, and significance assessed using Wilcoxon tests with Benjamini-Hochberg correction.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The sequencing data of phage Prima HS, the bacterial isolates and 16S rRNA analysis generated in this study have been deposited in the National Center for Biotechnology Information (NCBI) database under BioProject [PRJNA1275811](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1275811). The genomic data of Romulus used in this study is available at NCBI under accession code GCA_000907055.1 [https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000907055.1/]. Source data are provided in this paper.

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Author contributions

L.B., S.G., J.W., R.L., A.V.L., T.H., S.V., J.S., and J.M. conceptualized the study. Methodology was developed by L.B., S.G., A.V.L., J.W., R.L., S.V., J.S., T.H., J.M., D.D., I.S., Y.D., P.D.M., J.O., W.-J.M., V.V.N., and J.-P.P. Software development and computational analysis were performed by L.B. and D.J. The investigation was conducted by L.B., S.G., A.V.L., T.V., S.G., M.A., F.R., M.D., and M.M. Resources and clinical materials were

provided by T.V., A.V.L., D.D., R.L., J.W., J.O., M.M., and J.-P.P. Data curation was carried out by L.B., T.V., S.G., and D.J. Formal analysis and interpretation of results were performed by S.G., L.B., and M.M. Visualization of data and figures was completed by L.B., S.G., and M.M. The original draft of the manuscript was written by L.B. and S.G. All authors contributed to the review and editing of the manuscript. Supervision of the project was provided by R.L., A.V.L., and J.W. Project administration was coordinated by L.B., J.W., T.V., A.V.L., and R.L. Funding acquisition was led by J.W., R.L., A.V.L., T.H., S.V., J.S., and J.M.

Competing interests

The authors declare no competing interests.

Additional information

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